

武汉大学数学与统计学院 2025-2026 学年第一学期
《泛函分析》期中考试试题 (高力老师)

Note: we say two norms $\|\cdot\|$ and $\|\|\cdot\|\|$ are equivalent if there exist constants $c, C > 0$ such that for any x , $c\|x\| \leq \|\|x\|\| \leq C\|x\|$.

1. (20 points). Define the map T on $C([0, 1], \mathbb{R})$

$$T(f)(x) = \alpha + \int_0^x f(t)dt.$$

- (a) (5 points) State Fixed Point Theorem of Contraction Mapping.
(b) (5 points) For $M > 0$, define the following norm on $C([0, 1], \mathbb{R})$

$$\|f\| = \sup_{0 \leq x \leq 1} |f(x)|e^{-Mx}$$

Show that $\|\cdot\|$ is equivalent to the standard supremum norm $\|f\|_\infty := \sup_{0 \leq x \leq 1} |f(x)|$. This implies $C([0, 1], \mathbb{R})$ equipped with $\|\cdot\|$ is a Banach space.

- (c) (5 points) Find $M > 0$ such that T is a contraction with respect to the norm $\|\cdot\|$.
(d) (5 points) Based on (c), use Fixed Point Theorem to prove the following differential equation has unique solution.

$$f(0) = \alpha, f'(x) = f(x), x \in [0, 1].$$

2. (15 points).

- (a) (5 points) State Ascoli Theorem.

Let $C([0, 1])$ be the space of continuous functions on the unit interval and $K \in C([0, 1] \times [0, 1])$. Define the integral operator

$$(Tf)(x) := \int_0^1 K(x, y)f(y)dy, \quad f \in C([0, 1]), x \in [0, 1].$$

Let $\mathcal{B} := \{f \in C([0, 1]) : \|f\|_\infty \leq 1\}$ be the unit ball of $C([0, 1])$.

- (b) (5 points) Prove that T is bounded and calculate the norm of T .

- (c) (5 points) Use Ascoli Theorem to prove that $T(\mathcal{B})$ is relatively compact in $C([0, 1])$.

3. Problem 3 (20 points).

- (a) (5 points) State Banach-Steinhaus theorem (Uniform Boundedness Principle)

Let H be a Hilbert space and $u : H \rightarrow H$ be a linear map. Assume that

$$\langle u(x), y \rangle = \langle x, u(y) \rangle \quad \forall x, y \in H$$

- (b) (5 points) Prove that u is continuous.

Let $L_2([0, 2\pi])$ be the L_2 -space on the interval $[0, 2\pi]$. Let

$$C_{2\pi}^2 := \{f : [0, 2\pi] \rightarrow \mathbb{C} \mid f, f', f'' \text{ continuous and } f(0) = f(2\pi)\}$$

be the subspace of all periodic continuous twice differentiable function on $[0, 2\pi]$. Consider the Laplace operator

$$\Delta(f) = -f''$$

- (c) (5 points) Prove that for any $f, g \in C_{2\pi}^2$,

$$\langle \Delta(f), g \rangle = \langle f, \Delta(g) \rangle, \quad \langle \Delta(f), f \rangle \geq 0$$

- (d) (5 points) Prove that Δ is not bounded on $C_{2\pi}^2$ with respect to the L_2 -norm. Explain why this is not contradicting part (b).

4. Problem 4.

- (a) (5 points) Let H be a Hilbert space and let E be a subspace. State the theorem of orthogonal decomposition.

- (b) (20 points) Let E_1 and E_2 be two closed subspaces of H and let $E = E_1 \cap E_2$. Denote P_E , P_{E_1} and P_{E_2} as the orthogonal projections onto E , E_1 and E_2 respectively. Show that the following are equivalent:

- (i) $P_{E_1}P_{E_2} = P_{E_2}P_{E_1}$;

- (ii) $E^\perp \cap E_1$ is orthogonal to $E^\perp \cap E_2$;
- (iii) For any $x \in H$, $\|x - P_E(x)\|^2 \leq \|x - P_{E_1}(x)\|^2 + \|x - P_{E_2}(x)\|^2$;
- (iv) $P_{E_1}P_{E_2} = P_E$.

5. Problem 5 (20 points) The goal of this problem is to prove the following Ball-Pisier inequality: for $1 < p \leq 2$,

$$\left(\frac{\|f + g\|_p^p + \|f - g\|_p^p}{2} \right)^{\frac{1}{p}} \geq [\|f\|_p^2 + (p-1)\|g\|_p^2]^{\frac{1}{2}}, \quad \forall f, g \in L_p(\Omega, A, \mu).$$

where (Ω, A, μ) is a measure space. Note that for $p = 2$, this is the parallelogram rule.

(a) (5 points)

Prove that: for $a > b > 0$,

$$\left(\frac{(a+b)^p + (a-b)^p}{2} \right)^{1/p} \geq [a^2 + (p-1)b^2]^{1/2}.$$

Hint: define the functions φ and ψ on $[0, 1]$ by:

$$\psi(t) = \frac{(a+tb)^p + (a-tb)^p}{2}, \quad \varphi = \psi^{\frac{2}{p}}.$$

Show that:

$$\varphi(0) = a^2, \quad \varphi'(0) = 0, \quad \varphi''(t) \geq \frac{2}{p} \psi^{\frac{2}{p}-1}(t) \psi''(t) \geq 2(p-1)b^2.$$

(b) (5 points)

Prove that

$$\left(\frac{|z+w|^p + |z-w|^p}{2} \right)^{\frac{1}{p}} \geq (|z|^2 + (p-1)|w|^2)^{\frac{1}{2}}, \quad \forall z, w \in \mathbb{C}.$$

Deduce the Ball-Pisier inequality.

(c) (5 points)

Prove that the constant $p - 1$ is optimal for Ball-Pisier inequality, i.e., it cannot be replaced by a smaller constant.

Hint: it suffices to note that the constant $p - 1$ is optimal in the scalar inequality .

(d) (5 points)

Based on the proof above, conjecture the corresponding inequality for $2 < p < \infty$ and prove it.